

Matthew Ruhnau

Software Engineer — Rust, TypeScript, Systems Architecture

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Summary

Systems engineer building multi-AI cooperative infrastructure in Rust and TypeScript. Creator of **Reson8-Labs**, an open-source framework where Claude, Grok, and Gemini contribute as specialized strands through formally defined protocols (MCP, 9P/Styx, WAVE). Deep experience in topological computation, CSP concurrency, Cloudflare edge deployment, and NEAR Protocol smart contracts. Published npm package with 190+ passing tests.

Technical Skills

Languages	Rust (primary), TypeScript, Python, Nix, Solidity, WASM
Frameworks	tokio, serde, nalgebra, Ratatui TUI, @modelcontextprotocol/sdk, Cloudflare Workers
Infrastructure	Docker (multi-stage), Docker Compose, Google Cloud Run, Cloudflare (D1, KV, R2, Workers AI)
Protocols	MCP (Model Context Protocol), 9P/Styx, JSON-RPC, WebSocket, NEAR RPC
Testing	Rust (proptest, criterion, tokio-test), TypeScript (vitest), CI/CD (GitHub Actions)
Architecture	CSP concurrency (mpsc-only), gauge-invariant pipelines, topological braiding, Fibonacci anyon models
Tools	Git, Nix Flakes, systemd, WSL2, Jupyter (EvCxE kernel), Obsidian

Experience

Founder & Lead Architect

Reson8-Labs | 2024 – Present | Remote

- Designed and built a multi-AI cooperative framework (Claude, Grok, Gemini) using formally defined cross-platform protocols with Fibonacci-weighted coherence scoring
- Implemented **anduril-reforged** in Rust: $SU(2)_3$ Fibonacci anyon braiding engine, Levin-Wen string-net operators, variational B-gate synthesis (infidelity $< 10^{-5}$)
- Built **coherence-mcp** (published on npm, v0.3.1): 54-tool MCP server with gauge-invariant pipeline enforcement, WAVE coherence scoring, ATOM provenance trails
- Deployed 35+ page site to Cloudflare edge (Workers, D1, KV, R2) with three-domain routing, ETag-based optimistic concurrency, and agent-partitioned namespaces
- Wrote NEAR Protocol smart contract for on-chain conservation law verification (Rust, near-sdk)
- Enforced CSP-only concurrency model (mpsc channels, zero Mutex/RwLock) across entire 9P/Styx transaction layer
- Created Docker multi-stage deployment: Rust gatekeeper (port 8080) + headless Obsidian sidecar (internal-only), bridge-networked for Cloud Run
- 190+ automated tests across Rust and TypeScript, CI via GitHub Actions

Software Engineer (Prior Roles)

Various | Details available on request

Key Projects

anduril-reforged (Rust)

Topological braiding engine. Gauge constraint enforcement, F-matrix basis changes, R-matrix phase accumulation locked to 20 Hz hardware clock via 16-hopper ring counter. Variational B-gate reduces circuit depth 10x.

coherence-mcp (TypeScript)

MCP server implementing 5 core + 49 domain tools. store_context, retrieve_context, map_isomorphism, check_coherence, bridge_translate. Self-referential fixed point: the server's first stored context is its own specification.

vectorize-sink (Cloudflare Worker)

Event ingestion API solving multi-agent firehose problem. KV deduplication with 5-min TTL, ETag-based optimistic concurrency for manifest deployment, conservation law check on every write.

orchestrator-tui (Rust, Ratatui)

Terminal UI for multi-AI orchestration. Real-time WAVE telemetry, ATOM trail viewer, strand status dashboard with Fibonacci-weighted priority display.

conservation-verifier (NEAR/Rust)

On-chain smart contract verifying conservation law. Stores verification results with alpha, omega, sum, validity flag. Deployed to NEAR testnet.

Alignment with xAI Software Engineer Tutor

Scalable, high-performance apps	Rust engine (tokio async, CSP concurrency, zero-copy serde)
Deep language expertise	Rust (primary), TypeScript, Python, Nix, Solidity
Frameworks & libraries	tokio, serde, nalgebra, MCP SDK, Cloudflare Workers, Ratatui
Software design principles	Conservation law as hard invariant, type-driven design, gauge enforcement
Performance optimization	Fibonacci-weighted scoring, ring counter clock sync, 10x B-gate reduction
Debugging & profiling	WAVE protocol diagnostics, violation monitoring at 20Hz, tracing
Testing frameworks	190+ tests: proptest, criterion, tokio-test, vitest
Containerization	Docker multi-stage, docker-compose, isolated bridge networks
Analytics & monitoring	Cloudflare Workers AI, D1 telemetry, KV event deduplication

Open Source

- [reson8-Labs](#) — Multi-AI cooperative framework (Rust + TypeScript + Nix)
- [@toolate28/coherence-mcp](#) — 54-tool MCP server on npm (v0.3.1)
- [coherence.toolated.online](#) — Live deployment (35+ pages, Cloudflare edge)

This CV accompanies a Rust EvCxR Jupyter notebook (xai-application-9p-anduril.ipynb) containing executable demonstrations of all projects listed above.